MICROELECTROMECHANICAL SYSTEMS AND SENSORS

Course reader

Chapters 1–2 · draft for review

Bachelor programme · Electrical Engineering · 7th semester
Institute of Energy and Electrical Engineering
Draft, August 2026

How to use this reader. It does not replace a textbook. It carries the material that no textbook covers — the datasheet workflow, register-level interfacing, the specific parts on your bench — and points you at Morris & Langari, *Measurement and Instrumentation* (3rd ed., Elsevier 2020) for the underlying theory. Each chapter ends with exercises and full answers; work them before the corresponding laboratory.

Numerical values in worked examples have been checked by computation. Component figures labelled “Part A”, “Part B” and “Part C” are representative of real commercial devices but are not attributed to a named product; always verify against the datasheet revision in your hand.

1 · The measurement chain

What you should be able to do after this chapter

1. **Draw** the path from a physical quantity to a stored number, name each stage, and say what information can be lost at each one.
2. **Classify** an unfamiliar device as a sensor, a transducer or an actuator, and name the physical quantity it responds to.
3. **Predict**, from scaling arguments, why microscopic mechanical devices are fast and sensitive — and name at least one property that becomes *worse* as a device shrinks.
4. **Convert** a vague request into a measurement specification: quantity, range, resolution, accuracy, bandwidth, environment and output.

Notice that all four are things you *do*. None of them is "know about".

1.1 A motor that died under observation

A plant operated a 1500 rpm induction motor driving a pump. Bearing failure on that machine was expensive, so the maintenance department instrumented it: an accelerometer bolted to the bearing housing, a microcontroller, a daily log, and a trend line on a screen that an engineer actually looked at every week.

For six weeks the screen showed a vibration component near 20 Hz, slowly rising in amplitude. Everybody who saw it read it the same way: a gradual increase in load. Unremarkable.

In the seventh week the bearing seized and the motor was destroyed.

The post-mortem found no fault. The accelerometer met every number in its datasheet. The circuit board was sound. The firmware contained no bug. The logger had recorded every sample it was asked to record. The engineer had looked at the screen.

Every component met its specification, nobody was negligent, and the motor was scrap.

Hold that. We will return to it at the end of the chapter, once you have the equipment to explain it. The explanation is not exotic — it is arithmetic — and it is the reason this course exists.

1.2 Three words, used precisely

From here on these three words mean different things, and datasheets will not respect the distinction for you.

A **sensor** converts a physical quantity into a signal you can read. A MEMS microphone converts sound pressure into a voltage.

A **transducer** converts energy from one form into another. This is the general term; a sensor is the special case that points *inward*, from the world into your electronics. A loudspeaker, a piezoelectric disc and a strain gauge are all transducers.

An **actuator** converts a signal into a physical action on the world. A MEMS mirror converts a voltage into an angular deflection of a reflective surface.

Two remarks worth carrying forward.

First, **every actuator is also a measurement problem.** You only know that an actuator acted if you sense the result. A motor commanded to turn is not a motor that turned.

Second, some sensors are **passive**: they produce nothing at all on their own. A strain gauge changes its electrical resistance when it is stretched, but a resistance is not a signal. Until you pass a known current through it, or place it in a bridge with a known excitation voltage, it says nothing. Chapter 12 is largely about supplying that excitation correctly.

> **The quantity you are measuring has a name: the *measurand*.** Get into the habit of naming it explicitly and in SI units, because a surprising number of measurement failures begin with a team that never agreed on what it was measuring. "Vibration" is not a measurand. "Housing acceleration in m/s², RMS, in the band 1–4 kHz" is.

1.3 A measurement is a chain, not a component

Students arriving at this course usually picture a measurement as a component: you buy the sensor, you read the number. That picture is the single most expensive misconception in the field, and replacing it is the work of this chapter.

A measurement is a **path** from a physical quantity to a stored, trustworthy number. Every stage on that path transforms the information, and every stage can damage it. Figure 1.1 is the path. It is also, more or less, the syllabus of this course: every one of the sixteen lectures lives inside one of these seven boxes.

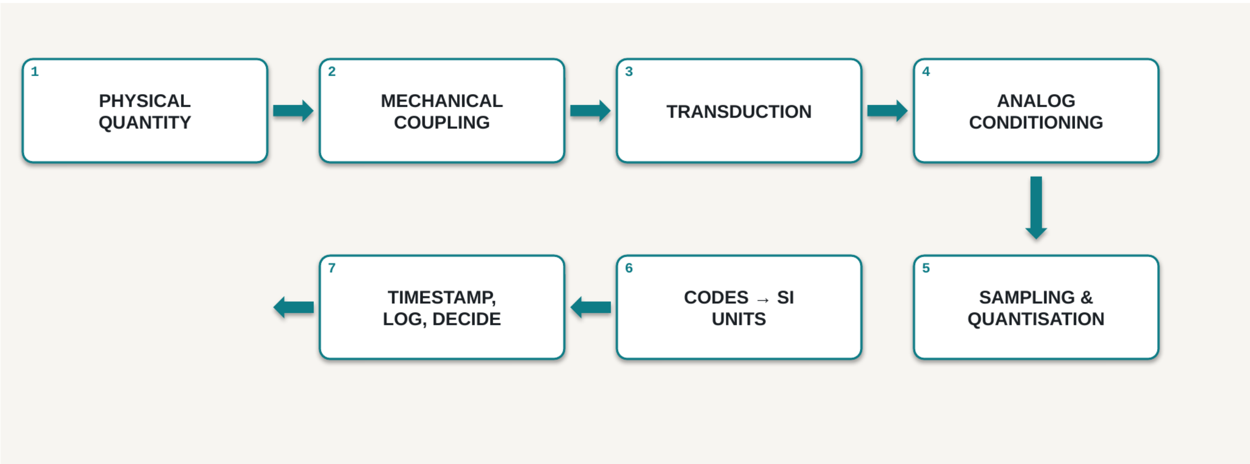

Figure 1.1 — The seven stages of a measurement chain. Read the top row left to right, then drop to the second row and read right to left.

Take the stages one at a time.

Stage 1 · The physical quantity. The only thing in the diagram that is unambiguously true. The motor really is vibrating. Everything to the right of this box is a story your instrument tells about it, and your job is to make the story faithful.

Stage 2 · Mechanical coupling. Before any electronics exist, the quantity has to reach the sensing element — through a bolt, a bracket, a housing, an adhesive pad, a length of tubing, a thermal interface. That path has its own transfer function: it attenuates, it delays, it resonates, and it does all three differently at different frequencies. It is part of your instrument, and it appears in no datasheet ever written. This is the stage that students omit and that experienced engineers check first.

Stage 3 · Transduction. Here physics becomes electricity: a capacitance changes, a resistance changes, a charge appears, a voltage develops. Chapter 4 covers the mechanisms; Chapters 5 to 11 cover which mechanism suits which measurand.

Stage 4 · Analog conditioning. Amplify, filter, level-shift, buffer. One filter in this stage — the **anti-alias filter** — is going to matter enormously by the end of this chapter. Remember that it lives here, in front of the converter.

Stage 5 · Sampling and quantisation. Two distinct operations that students merge for years, and they fail in different ways. **Sampling** chops *time*: the converter looks at the signal only at particular instants. **Quantisation** chops *amplitude*: each look is rounded to the nearest of a finite set of levels. Sampling too slowly and quantising too coarsely produce entirely different symptoms.

Stage 6 · Codes to SI units. A sensor never outputs an acceleration. It outputs an integer. Somebody multiplies that integer by a constant taken from a datasheet — and if the constant is wrong, everything downstream is confidently, precisely wrong. Chapter 2 is about where that constant comes from and what it depends on.

Stage 7 · Timestamp, log, decide. A sample without a time is nearly worthless. If you cannot say *when* a sample was taken, you cannot compute a frequency — and in the story that opened this chapter, frequency was the whole answer.

Figure 1.2 shows the same chain carrying one real vibration from housing to decision.

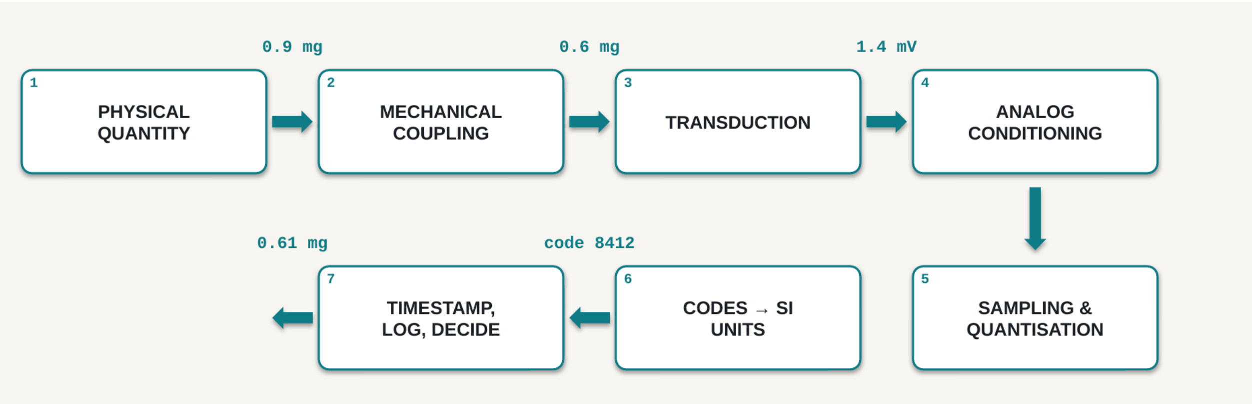

Figure 1.2 — One 0.9 mg vibration, all the way to a decision. Note stage 2: the mechanical path has already lost a third of the signal before a single electron has moved. Of all the numbers on this diagram, only those inside stages 3 to 6 would appear in any datasheet.

1.4 Where the truth dies, and whether you can recover it

Not all damage is equal. Some losses are annoying and can be repaired later by calculation or calibration; others are permanent and must be prevented by design. Knowing which is which is a large part of engineering judgement.

Stage	What is lost	Can you recover it?
2 Mechanical coupling	amplitude and phase, differently at each frequency	**No** — you must characterise it yourself
3 Transduction	cross-axis leakage, nonlinearity, temperature drift	Partly, by calibration
4 Conditioning	saturation clips peaks; a wrong filter removes the signal	**No**
5 Sampling	everything above half the sample rate, folded into your band	**Never**
5 Quantisation	detail below one least-significant bit	Partly, by averaging
6 Scaling to units	accuracy, if the sensitivity constant is wrong	Yes — recompute
7 Timestamping	the time axis, and therefore all frequency content	**No**

Table 1.1 — Losses along the chain. Three of these are permanent: they are design decisions, not calibration problems.

There is exactly one *never* in that table. Note which row it is on.

1.5 Why the microscopic world behaves differently

You have spent several years building mechanical intuition on machines you can hold. At the micrometre scale a good part of that intuition inverts, and it is worth knowing precisely which part.

What is actually inside

Figure 1.3 is a capacitive MEMS accelerometer in cross-section. A **proof mass** of mass m hangs on flexible suspension beams of stiffness k, between two fixed plates separated from it by gaps d. When the device accelerates, the mass lags behind its frame; one gap grows and the other shrinks; the difference in the two capacitances is the signal.

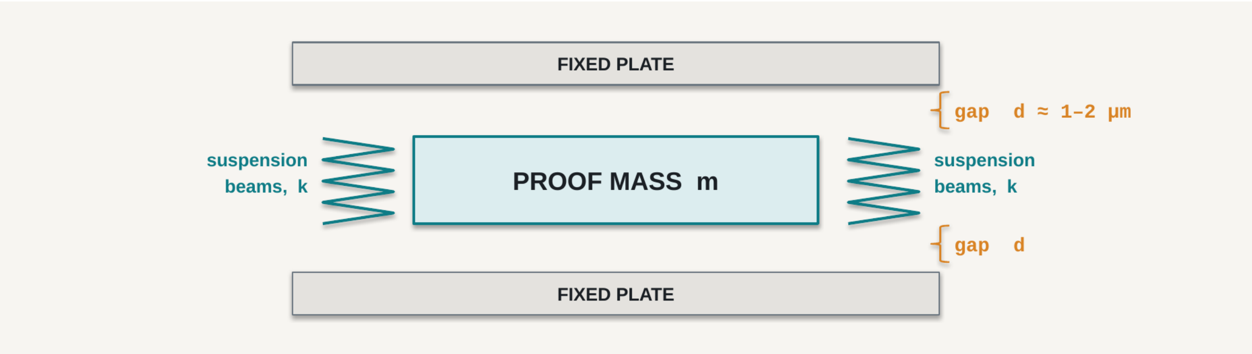

Figure 1.3 — A capacitive MEMS accelerometer. Acceleration displaces the proof mass; each gap changes; the differential capacitance $C = \varepsilon A/d$ is the output. Typical gaps are 1–2 µm — smaller than a red blood cell, which is why a single dust particle inside the package is a catastrophe and why Chapter 4 spends its time on packaging rather than on lithography.

If you understood the spring–mass system in mechanics, you already understand this device. Everything in a MEMS accelerometer datasheet is a consequence of *m*, *k* and *d*.

Scaling: the two exponents

Now shrink the whole structure. Scale every dimension — length, width, thickness — by the same factor *s*. This is called **isotropic scaling**, and it is the cleanest way to see what changes.

Mass follows volume:

$$m \propto L^3$$

Stiffness does not. For a beam of width *w*, thickness *t* and length *L*, the bending stiffness is

$$k = E \cdot w \cdot t^3 / 4L^3$$

Substitute *w*, *t* and *L* all scaled by *s*:

$$k \propto (s \cdot s^3) / s^3 = s$$

$$\text{so}$$

$$k \propto L^1$$

These two do not scale together, and that single fact is the whole of MEMS. Students assume the two effects cancel. They do not, and the gap between the exponents is where the entire field lives.

The consequence appears in the resonant frequency:

$$f_0 = (1 / 2\pi) \cdot \sqrt{(k / m)} \propto \sqrt{(L^1 / L^3)} = \sqrt{(L^{-2})} = 1 / L$$

Shrink a device by ten and its resonant frequency rises by ten.

Device	Size	Resonance
Bridge span	100 m	≈ 0.2 Hz
Tuning fork	10 cm	≈ 440 Hz
MEMS accelerometer	300 µm	≈ 5 kHz
MEMS gyroscope drive	100 µm	≈ 20 kHz
MEMS RF resonator	10 µm	≈ 100 MHz and above

Table 1.2 — Five orders of magnitude in size, eight in frequency.

This matters because a sensor cannot faithfully report signals near or above its own resonance. A high f_0 is not a curiosity; it is what buys you **bandwidth**, and bandwidth is what let anybody put an accelerometer on a bearing in the first place.

The other exponent, and the price you pay

Surface area also scales differently from volume:

$$A / V \propto L^2 / L^3 = 1 / L$$

As things get smaller, they become almost entirely surface. Gravity and inertia are volume effects; friction, adhesion and surface tension are area effects. Shrink far enough and the area effects win — which is why a microscopic mechanical structure can touch itself and simply stay stuck. That failure is called **stiction**, and it is a real production concern.

So the honest ledger of shrinking a device looks like this:

Gets better	Gets worse
Bandwidth, since $f_0 \propto 1/L$	Thermomechanical noise floor
Response time, thermal settling	Stiction; surface forces dominate
Power per device	Sensitivity to packaging stress
Cost per device at volume	Temperature drift and offset stability
Thousands per wafer, batch fabricated	Contamination: a dust speck is fatal

Table 1.3 — Shrinking is a trade, not an improvement.

The noise entry deserves a sentence, because it is the honest answer to *"why not just make them smaller?"*. The proof mass is what averages out the random thermal buffeting of the gas molecules around it. The noise-equivalent acceleration produced by that buffeting scales roughly as

$$a_n \propto \sqrt{4 k_B T \omega_0 / m Q}$$

so it **rises as the proof mass falls**. You buy bandwidth and you pay in noise floor. The datasheet will make you choose, and Chapter 2 is where you learn to read the price.

> **A macroscopic engineer worries about mass. A MEMS engineer worries about surfaces.**

1.6 Worked calculation: how a fast signal becomes a slow one

We can now explain the motor.

The **Nyquist–Shannon sampling theorem** states that to represent a signal containing frequencies up to f_{max}, you must sample at a rate

$$f_s > 2 f_{max}$$

Sample more slowly and the information is not merely missing — it is *misfiled*. A component at frequency f higher than $f_s/2$ reappears inside your measurement band at the **alias frequency**

$$f_{alias} = | f - n \cdot f_s | \quad (n = \text{the nearest integer multiple})$$

It arrives indistinguishable from a real signal at that lower frequency. Nothing downstream can separate them, because by the time the samples exist the distinction has already been destroyed.

The numbers from the post-mortem

Three facts were established after the motor failed:

- the bearing-fault signature sat at **1520 Hz**
- the system sampled at **100 Hz**
- there was **no anti-alias filter** in stage 4

To observe 1520 Hz honestly you would need to sample above 3040 Hz. The system sampled thirty times too slowly. So where did that energy go?

$$n = \mathrm{round}(\,1520 \,/\, 100\,) = 15$$

$$f_{alias} = |\,1520 - 15 \times 100\,| = |\,1520 - 1500\,| = 20\ \mathrm{Hz}$$

There is the 20 Hz line. It was never load variation. It was the bearing destroying itself, faithfully recorded and filed under the wrong frequency.

Figure 1.4 shows the mechanism. The fast teal signal is the real vibration. The red dots are the only instants at which the converter looked. Draw the smoothest curve you can through just those dots and you get the dashed red line — a slow oscillation that is not present in the physical world at all.

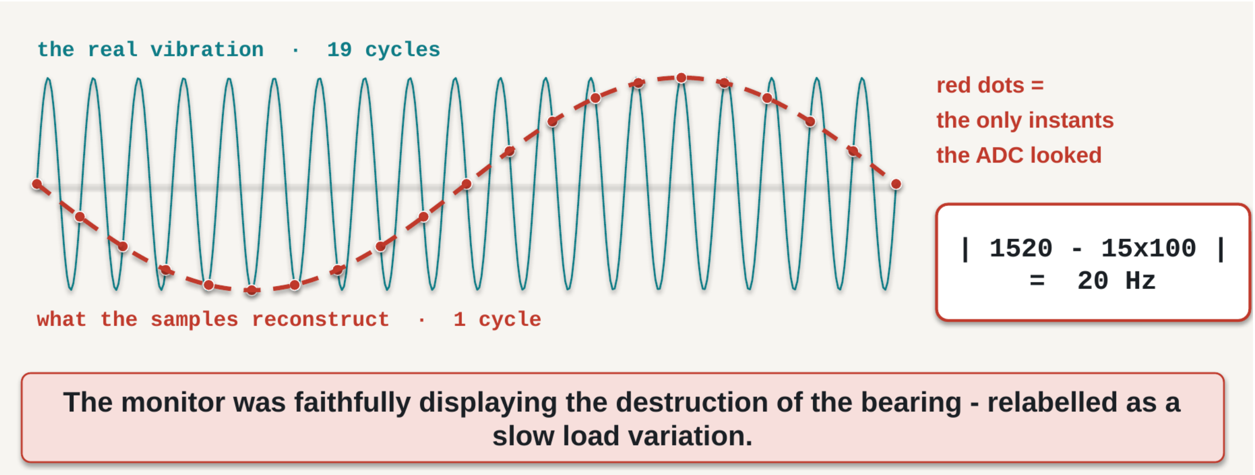

Figure 1.4 — Aliasing. Drawn at 19 signal cycles per 20 samples so the mechanism is visible; the motor's real ratio was 1520:100, which is identical arithmetic. The samples are perfectly accurate. Their *interpretation* is not.

Why nobody caught it

Here is the detail that turns this from a curiosity into a lesson. The shaft turned at 1500 rpm, which is **25 Hz**. A vibration component at 20 Hz was therefore entirely plausible to every engineer who looked at that screen — the same order as the running speed, drifting slowly upward, exactly what a gradually loading machine does.

The wrong answer was believable. That is why it survived six weeks of weekly review.

Two conclusions, and they are the reason this chapter exists:

1. **Aliasing is permanent.** No filter applied afterwards, no clever software, no amount of machine learning recovers a frequency you failed to sample. The only defence is an anti-alias filter in stage 4 and an adequate rate in stage 5 — both chosen *before* the system is built.
2. **Nothing was broken. Every part met its specification. The system was never specified.** No individual component owned the sampling decision, so nobody made it.

1.7 From a wish to a specification

Nobody will hand you a specification. They will hand you a sentence like:

> *"Just tell me if the motor is vibrating too much."*

There is not one number in it. Until there is, you cannot buy anything, you cannot design anything, and — importantly — you cannot be held to anything. Converting that sentence into an engineering document is the first task of every measurement project, and these seven questions do it.

	Question
Quantity	What physical variable, in what units?
Range	The smallest and largest value that must be reported?
Resolution	The smallest change that must be distinguishable?
Accuracy	How wrong may a reading be — **and over what temperature range**?
Bandwidth	How fast does it change? What is the highest frequency that matters?
Environment	Temperature, humidity, vibration, EMI, supply, power budget?
Output	Who consumes the number, in what form, how often, timestamped how?

Table 1.4 — Seven questions that turn a wish into a specification.

Two of these deserve emphasis. **Accuracy** without a temperature range is marketing, not engineering; Chapter 2 shows exactly how much damage that omission does. And **bandwidth** is the question that killed the motor.

Applied to the motor, the wish becomes this:

	Specification	Where the number came from
Quantity	housing acceleration, m/s² RMS	vibration is what a failing bearing radiates
Range	±16 g	impact transients, not steady vibration
Resolution	≤ 5 mg	earliest detectable fault ≈ 30 mg
Accuracy	±5 % of reading, 0–70 °C	trend detection, not absolute metrology
Bandwidth	**≥ 4 kHz, anti-aliased**	bearing signature spans 1–4 kHz
Environment	0–70 °C, 24 V rail, oily, EMI from a drive	the motor's actual cabinet
Output	RMS + spectrum, 1 Hz, timestamped ±1 ms	so that a frequency can be computed

Table 1.5 — The same request, specified.

Notice two things about Table 1.5. Every line has a **reason** — a specification value without a justification is a guess wearing a suit. And notice what is absent: no manufacturer, no part number, no price. Those come afterwards, and choosing them is the subject of Chapter 2.

The bandwidth line is the one that was missing from the real system. One line, in one document:

> `Bandwidth: ≥ 4 kHz, anti-aliased. Sample rate: ≥ 8 kHz.`

One line. One motor.

1.8 Reading

This chapter's material corresponds to **Morris & Langari, *Measurement and Instrumentation*, 3rd edition (Elsevier, 2020), Chapter 1**, "Fundamentals of Measurement Systems". Read it for the formal treatment of the measurement-system block diagram and the vocabulary of measurement.

The sampling theorem and quantisation are developed properly in **Chapter 6**, "Data Acquisition and Signal Processing" — we return to that material in Chapter 3 of this reader, so a first pass now is enough.

For MEMS structures and fabrication, the free ITMO text *Микроэлектромеханические системы и датчики* (Университет ИТМО, 2020) covers device constructions and operating principles in its Chapters 1 and 2, with self-check questions. We use it properly in Chapter 4.

Chapter and section numbering should be checked against the edition available to you.

1.9 Exercises

1.1 List the stages of a measurement chain in order, from physical quantity to stored value.

1.2 Give one example of information lost at the analog-to-digital converter that cannot be recovered by any later processing. Explain why "cannot" is the right word.

1.3 A MEMS resonator is scaled down isotropically by a factor of 5. What happens to its resonant frequency, and why?

1.4 Name one sensor property that gets *worse* as a device shrinks, and explain the mechanism.

1.5 Classify each of the following, and name its measurand or its output action: (a) a MEMS microphone, (b) a piezoelectric buzzer, (c) a strain gauge, (d) a MEMS mirror. Which one produces no signal at all without external excitation?

1.6 A system samples at 500 Hz with no anti-alias filter. A real 1800 Hz component is present. At what frequency does it appear in the recorded data?

1.7 A drone must hold its altitude to ±0.5 m using a barometric pressure sensor. (a) What is the measurand — and is it altitude? (b) Roughly what pressure change corresponds to 0.5 m of altitude near sea level? (c) Explain why the *range* and the *resolution* required of this sensor differ by several orders of magnitude.

1.8 In one sentence: why did the motor's monitor display a calm 20 Hz line?

1.9 Write the seven specification lines for the following request: *"The greenhouse gets too hot in the afternoon — do something about it."* State an assumption wherever the request does not give you enough information.

Answers

1.1 Measurand → mechanical coupling and mounting → transduction → analog conditioning, including anti-alias filtering → sampling and quantisation → raw codes scaled to SI units → timestamp, log, decide.

1.2 Any frequency content above half the sample rate. Sampling folds it into the measurement band as an alias, arriving indistinguishable from a genuine low-frequency component; since no record of the original frequency survives in the samples, no later processing can separate them. Detail below one LSB is also lost, though averaging can partially recover it.

1.3 It rises by a factor of 5. Under isotropic scaling $k \propto L$ while $m \propto L^3$, so $f_0 \propto \sqrt{(k/m)} \propto 1/L$.

1.4 The thermomechanical noise floor: noise-equivalent acceleration rises as the proof mass falls, so a smaller device has a worse noise floor. Alternatively stiction, because $A/V \propto 1/L$ means surface forces come to dominate body forces; or increased sensitivity to package stress and thermal drift.

1.5 (a) Sensor; measurand sound pressure. (b) Actuator (and a transducer); converts an electrical signal into acoustic output. (c) Sensor; measurand strain — **this is the passive one**, since it only changes resistance and needs an excitation current or bridge voltage to produce a signal. (d) Actuator; converts a voltage into angular deflection of a mirror.

1.6 n = round(1800/500) = 4, so $f$alias = |1800 − 2000| = **200 Hz**.

1.7 (a) The measurand is **absolute air pressure**, in pascals or hectopascals; altitude is a *derived* quantity, computed from pressure using a model of the atmosphere. (b) Near sea level pressure falls by roughly 12 Pa per metre, so 0.5 m ≈ **6 Pa**, about 0.06 hPa. (c) The sensor must *operate over* the full range of weather and site pressure — perhaps 950 to 1050 hPa, a span of 10 000 Pa — while *resolving* changes of about 6 Pa. Range and resolution therefore differ by more than three orders of magnitude, and confusing the two is one of the most common specification errors.

1.8 The ≈1520 Hz bearing-fault energy was aliased by a 100 Hz sample rate with no anti-alias filter, appearing at |1520 − 15×100| = 20 Hz, which was misread as normal load variation because the shaft itself turned at 25 Hz.

1.9 Answers will vary; a defensible version is: **Quantity** air temperature in °C at canopy height (assumption: air, not soil or leaf temperature). **Range** −10 to +60 °C. **Resolution** 0.5 °C. **Accuracy** ±1 °C over 0–50 °C. **Bandwidth** very low — a greenhouse has a thermal time constant of minutes, so 0.01 Hz is ample, and one sample per minute is generous. **Environment** high humidity, condensation, direct sunlight requiring a radiation shield, mains supply available. **Output** one timestamped reading per minute to a controller, with a threshold that opens a vent. Credit belongs to any answer that states its assumptions and justifies its numbers — particularly one that notices the bandwidth is *low*, and that direct sunlight on the sensor is the dominant error source rather than the sensor itself.

2 · Specifications, and choosing a part

What you should be able to do after this chapter

1. **Distinguish** accuracy, precision, resolution and sensitivity, and sort a list of parameters into static and dynamic characteristics.
2. **Locate** a number in a real datasheet together with the conditions under which it was measured, and state at least one performance limit the datasheet cannot specify.
3. **Compute** noise-limited resolution from a noise-density figure and a bandwidth, and decide whether noise or quantisation dominates.
4. **Select** between candidate devices by building an error budget, and defend the choice by naming the dominant error term.

2.1 Two front pages

You have written a specification. Now you must buy something. Thousands of parts exist, every number is published, and the datasheets are free — so selection looks like a table lookup.

Here is the job:

> A solar-tracker frame must report its tilt angle to **±0.5°**, outdoors, over an ambient range of **0 to 40 °C**. It will be calibrated once on the bench, at 20 °C.

And here are two candidates, as their front pages present them.

PART A	
Accelerometer, 3-axis	
Resolution	`16-bit`
Full scale	`±2 g`
Sensitivity	`0.061 mg/LSB`
Noise density	`100 µg/√Hz`
Unit price	`€1.80`

"HIGH RESOLUTION · LOW NOISE"

PART B	
Accelerometer, 3-axis	
Resolution	`14-bit`
Full scale	`±4 g`
Sensitivity	`0.488 mg/LSB`
Noise density	`100 µg/√Hz`
Unit price	`€4.20`

The job: report the tilt of a solar-tracker frame to ±0.5°, outdoors, 0 to 40 °C.

Figure 2.1 — Two candidate accelerometers as advertised. Part A offers eight times finer resolution at less than half the price, and says so. Part B says nothing for itself. *(These are representative commercial figures rather than a named product, chosen so that the arithmetic in this chapter is exact.)*

Which do you specify? Commit to an answer now, before reading on.

Most engineers choose Part A, and most of the time the reasoning is not lazy — sixteen bits genuinely is better than fourteen, and €1.80 genuinely is better than €4.20. By the end of this chapter you will be able to show that **Part A cannot meet the requirement**, that **Part B meets it with a fourfold margin**, and that the number which decides it appears on neither front page.

2.2 First, turn the requirement into a number

You cannot budget an error against an angle. So before reading any datasheet, convert the requirement into the physical quantity the sensor actually measures.

A tilted accelerometer at rest senses a component of gravity. For a tilt of angle θ from horizontal, the component along the sensitive axis is g sin θ. So a tilt of half a degree corresponds to

```
sin(0.5°) × 1 g = 0.008727 g = 8.73 mg
```

Write that number down and keep it in front of you for the rest of the chapter.

> **8.73 mg is the entire signal we are trying to measure.**

Every error term from here on gets compared against those 8.73 mg. Any term larger than that is fatal. Any term far below it is irrelevant — no matter how impressive it looks on a front page. This single conversion turns *"which is better?"*, which is an opinion, into arithmetic.

2.3 Four words that get confused

Most engineering arguments about sensors are really arguments about these four words.

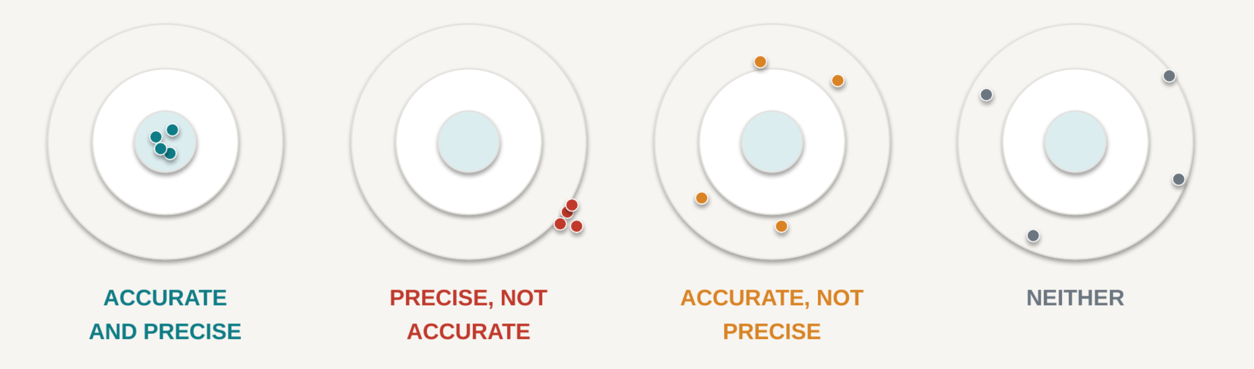

Figure 2.2 — The same target, four different instruments. The centre is the *true* value.

Precision is how tightly the shots group. **Accuracy** is where the group sits relative to the truth. They are independent properties: the second target — tightly grouped and completely wrong — is the dangerous instrument, because every reading agrees with every other reading and they are all wrong by the same amount.

Resolution does not appear on that picture at all. Resolution is how finely you can *report* where the shot landed. You can report a wrong position to six decimal places, and this is precisely the trap Part A is setting.

Sensitivity is different again: it is the change in output per unit change in input — for our accelerometer, mg per least-significant bit. Sensitivity tells you how the scale is drawn, not how truthful it is.

Consider a concrete case. A pressure sensor sits in a chamber held at a true, constant 1000.0 hPa. It is read one hundred times. Every reading falls between 1012.3 and 1012.5 hPa.

That device is **precise but not accurate**: superbly repeatable, and 12.4 hPa wrong. And it raises the question that carries the rest of this chapter:

> **Which of these two problems can a calibration fix?**

The 12.4 hPa offset — **yes**. Measure it once against a reference and subtract it.

The 0.2 hPa scatter — **no**. Random scatter cannot be removed by calibration, only reduced by averaging, and averaging costs you bandwidth. If you average sixteen samples to cut the noise by a factor of four, you have also made your instrument sixteen times slower.

That gives us a rule worth memorising, because it is the spine of Chapter 14 and of several laboratories:

> **Systematic error is a calibration problem. Random error is a bandwidth problem. Drift is a component-selection problem.**

The third line is the one that ruins projects. Drift passes every test you run on the bench at 22 °C and fails in the field in February, because it *moves after you calibrated it*. You cannot calibrate away a quantity that changes when you are not looking.

2.4 The vocabulary, sorted

Datasheets are organised around this vocabulary, so it is worth having sorted once. **Static** characteristics describe behaviour when nothing is changing; **dynamic** characteristics describe behaviour when things change.

Static	Meaning
Range	smallest to largest measurable value
Sensitivity	output change per unit of input
Resolution	smallest distinguishable change
Threshold	smallest input that produces any output at all
Offset	output when the input is zero
Linearity	deviation from a straight-line response
Hysteresis	does the reading depend on which direction you approached from?
Repeatability	same input, same output, measured again later

Dynamic	Meaning
Bandwidth	highest frequency reported faithfully
Response time	time to settle after a step input
Output data rate (ODR)	samples delivered per second — **not** the same as bandwidth
Noise density	noise per √Hz; becomes an RMS figure once you choose a bandwidth
Group delay	how late the filtered answer arrives
Cross-sensitivity	response to the axis or quantity you did *not* want
Drift	slow change with time and temperature

Table 2.1 — Static and dynamic characteristics. Two entries are worth flagging now.

ODR is not bandwidth. A sensor will happily hand you two hundred numbers per second while telling you nothing about signal content above 50 Hz. Confusing the two is the most common arithmetic error in this chapter, and §2.6 shows exactly what it costs.

Cross-sensitivity is real in every device: an accelerometer shaken along *x* reports a little *z*. We return to it in §2.8 for an important reason.

2.5 Read the conditions first, then the number

A datasheet is a legal document, not a promise. The headline is on page one; the truth is in the conditions.

Parameter	Min	Typ	Max	Unit	Conditions
Sensitivity	–	`0.061`	–	mg/LSB	FS = ±2 g, 25 °C
Zero-g level	–	`±40`	–	mg	25 °C, after soldering
Zero-g temp. coefficient	–	`±0.5`	–	mg/°C	−40 to +85 °C
Noise density	–	`100`	–	µg/√Hz	ODR = 200 Hz, BW = 50 Hz
Cross-axis sensitivity	–	`±1`	–	%	package level, not board level

▸ **Every number is “typ”.** No min, no max. You are being told the average of a production run, not a

Figure 2.3 — A representative extract from an accelerometer datasheet. The rightmost column is the one to read first.

Two observations about Figure 2.3 that generalise to almost every sensor datasheet you will meet.

Every number is "typ". There is no minimum and no maximum. You are being told the *average of a production run*, not a guarantee. If your design works only with a typical part, then roughly half of your production does not work — and you will discover this at volume, which is the worst possible time.

The conditions differ from row to row. Sensitivity is quoted at 25 °C. The temperature coefficient is quoted over a 125 °C span. Noise density is quoted at one specific bandwidth. **As printed, these rows are not comparable with one another**, and they are certainly not comparable with the corresponding rows of a competitor's datasheet until you have checked that the conditions match.

So read a datasheet right to left: **conditions first, parameter name second, and the number last.** Most people read it in exactly the opposite order, and that is how they get surprised in February.

2.6 Worked calculation: the error budget

We now have everything needed to decide between Part A and Part B. There are four terms.

Term 1 · Noise

A noise-density figure is not yet a noise figure. It becomes one only when you choose a bandwidth:

```
noise_RMS = noise density × √bandwidth
```

We are measuring a static orientation, so we can afford a modest bandwidth. Take 50 Hz. Both parts specify 100 µg/√Hz:

```
100 µg/√Hz × √(50 Hz) = 100 × 7.07 = 707 µg = 0.707 mg
```

Check the units, because unit cancellation — not the physics — is what usually goes wrong here: µg/√Hz × √Hz = µg. If your units do not cancel, you have used the wrong number.

Note carefully what we did *not* use. The datasheet quoted this figure at an ODR of 200 Hz with a bandwidth of 50 Hz. Using √200 would give 1.41 mg — a wrong answer that looks entirely reasonable. **Bandwidth, not ODR.**

There is a useful corollary. Noise is not a property of the sensor alone; it is a property of the sensor *and the bandwidth you chose*. Halve your bandwidth and you improve noise by √2, for free, if the signal allows it. Widen it to chase fast events and you pay in noise. This is the same trade the motor's monitor got wrong in Chapter 1, seen from the other side.

Term 2 · Quantisation

Rounding to the nearest code contributes an RMS error of one least-significant bit divided by √12:

```
Part A: 0.061 / 3.464 = 0.018 mg Part B: 0.488 / 3.464 = 0.141 mg
```

Both are negligible against 8.73 mg. Note what this means: Part A's celebrated eight-times-finer resolution buys it an advantage of about 0.12 mg on a term that was already invisible.

Term 3 · Offset

Part A's zero-*g* offset is ±40 mg; Part B's is ±10 mg. Both are far larger than our 8.73 mg measurand — and both **go to zero**, because we calibrate the instrument once on the bench at 20 °C and subtract what we find. This is the calibration column, and it is why a large raw offset need not disqualify a part.

Term 4 · Offset drift

Here is the term nobody computed.

The offset does not stay where you calibrated it. It moves with temperature, at a rate given by the temperature coefficient. We calibrate at 20 °C and operate from 0 to 40 °C, so the worst-case excursion is $\Delta T = \pm 20$ °C:

$$\text{Part A: } 0.5\ \text{mg/°C} \times 20\ \text{°C} = 10.0\ \text{mg} \quad \text{Part B: } 0.1 \times 20 = 2.0\ \text{mg}$$

Part A drifts by 10 mg. The entire quantity we are trying to measure is 8.73 mg. The error is larger than the signal, and it appears *after* the calibration, so no bench procedure catches it.

Combining the terms

These four error sources are independent, so they add in quadrature — root-sum-square:

$$\varepsilon_{total} = \sqrt{\varepsilon_{noise}^2 + \varepsilon_{quant}^2 + \varepsilon_{drift}^2}$$

Term	Part A	Part B
Noise, 100 µg/√Hz × √50 Hz	0.707 mg	0.707 mg
Quantisation, LSB/√12	0.018 mg	0.141 mg
Offset drift, TC × ΔT	**10.000 mg**	**2.000 mg**
Total (RSS)	**10.02 mg**	**2.13 mg**
As a tilt angle, arcsin(mg/1000)	**0.574°**	**0.122°**
Against the ±0.5° requirement	**FAILS**	**passes, 4× margin**

Table 2.2 — The error budget. Bandwidth 50 Hz, calibrated at 20 °C, operated over 0–40 °C.

Look at what root-sum-square does. Combining 10 mg with 0.707 mg gives 10.02 mg: the noise term we spent half a page computing changed the answer by three hundredths of a milli-*g*.

> **In any error budget, find the largest term first. Everything else is decoration.**

The verdict

Part A resolves eight times finer than Part B and cannot do the job. Part B's 0.488 mg/LSB already resolves 0.028° of tilt — eighteen times finer than the half-degree we need — so resolution was never the deciding variable. It only looked like it was, because it was the number printed on the front page.

And note the incentives. Part A was cheaper, had finer resolution, and carried a marketing line. Every visible signal pointed at the part that fails.

If you chose "I cannot decide from the front pages alone" at the start of this chapter, you were right, and right for the best possible reason. If you chose Part A, you did what most engineers do. If you now choose Part B, you can defend it with a number — which is the only defence that survives a design review.

2.7 The part on your bench

Parts A and B were built to make the arithmetic clean. Now do it for real, on the device you will wire up in the laboratory: the **ST ISM330DHCXTR**, an industrial-grade 6-axis inertial module. Here is what its datasheet actually says about the accelerometer.

Parameter	Min	Typ	Max	Unit
Sensitivity, ±2 g full scale		0.061		mg/LSB
Sensitivity tolerance	−2		+2	%
Acceleration noise density, high-performance mode		**60**	**100**	µg/√Hz
Zero-g level offset accuracy	−65	**±10**	+65	mg
Zero-g level change vs. temperature	−0.5	**±0.1**	+0.5	mg/°C
Sensitivity change vs. temperature, −40 to +105 °C	−0.01	±0.005	+0.01	%/°C
Cross-axis sensitivity, **at T = 25 °C**		±0.5		%
Operating temperature range	−40		+105	°C

Table 2.3 — ISM330DHCX accelerometer characteristics, from datasheet DS13012 Rev 6. Note that the two columns that decided our design — noise density and zero-g temperature drift — each carry both a typical *and* a maximum value.

Now run the same error budget twice: once with the typical column, once with the maximum. Bandwidth 50 Hz, calibrated at 20 °C, operated 0–40 °C, so ΔT = ±20 °C, exactly as before.

Term	Using typ	Using max
Noise, density × √50 Hz	0.424 mg	0.707 mg
Quantisation, 0.061/√12	0.018 mg	0.018 mg
Offset drift, TC × 20 °C	**2.000 mg**	**10.000 mg**
Total (RSS)	**2.05 mg**	**10.02 mg**
As a tilt angle	**0.117°**	**0.574°**
Against the ±0.5° requirement	**passes, 4× margin**	**FAILS**

Table 2.4 — The same part, the same requirement, the same arithmetic. One datasheet column apart.

Read that again, because it is the most important table in this chapter.

This is one device. It is not a comparison between a good part and a bad one. A typical ISM330DHCX meets our tilt requirement comfortably; a part at the limit of its own published specification **misses it** — and both parts are entirely within specification, both would pass incoming inspection, and both would be shipped to you in the same reel.

So what do you actually do?

- **If you must guarantee the specification**, you budget with the **max** column. On that basis this design does not close, and you need a different approach: tighten the temperature range, measure temperature and compensate the offset (Chapter 14), or relax the ±0.5° requirement.
- **If you are building ten units** and can measure each one, the typical column plus a per-unit calibration is a defensible engineering decision — provided you write down that you made it.
- **What you must not do** is budget with `typ` and describe the result as a guarantee. That is the difference between an engineering document and a hope.

Notice also what our fictional parts were really doing. Part A's figures (±40 mg offset, ±0.5 mg/°C, 100 µg/√Hz) were essentially this real part's **worst case**, and Part B's (±10 mg, ±0.1 mg/°C) were its **typical case**. The lesson has not changed; it has just become harder to dismiss, because it is now a single part number rather than a choice between two.

> **A datasheet does not describe your device. It describes the population your device was drawn from.**

2.8 What the datasheet cannot tell you

Look again at the cross-axis row of Table 2.3. The ISM330DHCX quotes ±0.5 % — a real, published number. Now ask a slightly different question:

> **What is this device's cross-axis sensitivity after it has been soldered onto my board?**

That is not there, and it cannot be. Look at the condition attached to the published figure: **T = 25 °C**, for the packaged die, as measured by the manufacturer. But board-level mounting stress, solder-joint asymmetry, thermal gradients across the PCB and flexing of the board all change it — and the manufacturer cannot know what your board does to their part.

The honest engineering answer is therefore in the first person: *the datasheet cannot tell me this; I must measure it on my own assembled board.* Installing that habit is the most important thing this chapter does, and it is why the laboratory programme includes characterisation work rather than only bring-up work.

The same applies to the mechanical coupling of Chapter 1, to the thermal environment inside your enclosure, and to the actual noise you achieve after routing a sensitive analog trace next to a switching regulator. **The datasheet describes a component. You are building a system.**

2.9 Selecting, and over-selecting

Selection is a calculation against a specification, documented so that somebody else can check it. A simple criteria table is enough, provided the weights are honest.

Consider a third candidate. **Part C**: 16-bit, ±2 g, 60 µg/√Hz, offset ±5 mg, temperature coefficient ±0.05 mg/°C, 0.9 mA, €11.50.

Criterion	Weight	A	B	C
Meets ±0.5° over 0–40 °C	pass/fail	fail	pass	pass
Total error as an angle	high	0.574°	0.122°	0.06°
Unit price at 500 off	medium	€1.80	€4.20	€11.50
Current draw	medium	0.15 mA	0.18 mA	0.90 mA
A library already exists	**low**	yes	no	no

Table 2.5 — A criteria table. Note the weight on the last row.

Part C is technically the best part in the table. It is also **the wrong answer**, and for an instructive reason: at 500 units it costs €3 650 more than Part B, draws five times the current, and buys no capability that the requirement asks for. Over-specification is an engineering failure too — a quieter one than under-specification, but you still have to explain it.

Which brings us to the last row, and to a rule this course applies throughout:

> **Never choose a component only because a software library exists for it.**

A library is a genuine asset and it may save you a week. It is not a specification. Rank it last, not zero. Part A had a library; Part A cannot do the job.

Choose instead on the features that expose the engineering you need to do: selectable range and output data rate, documented bandwidth and noise, a self-test function, a data-ready interrupt, a FIFO where relevant, an accessible register map, and a complete datasheet.

2.10 Reading

The core of this chapter corresponds to **Morris & Langari, *Measurement and Instrumentation*, 3rd edition (Elsevier, 2020), Chapter 2**, "Instrument Types and Performance Characteristics". Read it for the formal definitions of the static and dynamic characteristics in Table 2.1, and for its worked examples.

Chapter 3 ("Measurement Uncertainty") and **Chapter 4** ("Statistical Analysis of Measurements Subject to Random Errors") develop error combination — including the root-sum-square rule used in Table 2.2 — much more fully than we have here. We return to that material in Chapter 14; a first reading now will make the error budget above feel less like a recipe.

Then read the electrical-characteristics section of the **ISM330DHCX datasheet (DS13012)** — the part on your own bench. That is not an optional extra: from this chapter onward every laboratory assumes you can find a number, and its conditions, in a real document. Table 2.3 above is where you should end up; arriving there yourself is the skill.

Chapter and section numbering should be checked against the edition available to you.

2.11 Exercises

2.1 Define accuracy, precision and resolution so that the three definitions cannot be confused with one another. Give an example of a device that is precise but inaccurate.

2.2 Which of the following can a single-point bench calibration remove: offset, random noise, temperature drift, quantisation? Explain each answer.

2.3 A sensor specifies 150 µg/√Hz. You need 20 Hz of bandwidth. (a) Compute the RMS noise. (b) State what happens if you widen the bandwidth to 80 Hz, and by what factor.

2.4 A datasheet gives an offset temperature coefficient of ±0.3 mg/°C. The device is calibrated at 25 °C and used from −10 to +60 °C. Compute the worst-case offset error from drift alone, and express it as a tilt angle.

2.5 Your requirement is a signal of 8.73 mg. A candidate part offers 0.061 mg resolution and 10 mg of offset drift over your temperature range. In one sentence, explain what is wrong with calling this a "high-resolution solution".

2.6 A sensor is quoted as "±0.05 % of full scale" accuracy at 25 °C, with no temperature coefficient given anywhere in the datasheet. Full scale is 20 kPa. (a) What is the accuracy in pascals at 25 °C? (b) Why can you *not* use this part in an error budget for outdoor operation, even though its quoted accuracy is excellent?

2.7 Give one performance-limiting property that a datasheet cannot specify for your finished product, and say who must determine it.

2.8 You must measure water level in an outdoor tank 2 m deep, to ±20 mm, with water between 5 and 35 °C. Sensor P has a total error band of ±0.25 % of full scale over 0–50 °C. Sensor Q is specified as ±0.05 % of full scale at 25 °C, with no temperature coefficient stated. Full scale for both is 20 kPa. Which do you specify, and why? (1 mm of water ≈ 9.81 Pa.)

2.9 Using Table 2.3, compute the ISM330DHCX's RMS noise at a bandwidth of 200 Hz, for both the typical and the maximum noise density. Express each as a tilt angle.

2.10 The ISM330DHCX's sensitivity tolerance is ±2 %. You measure 1 g of gravity and the device reports 1015 mg. (a) Is this part within specification on sensitivity? (b) Which single bench measurement would let you correct this error, and what would remain uncorrected afterwards?

2.11 An accelerometer offers selectable ranges of ±2 g, ±4 g, ±8 g and ±16 g at a fixed 14-bit output. (a) Compute the resolution in mg/LSB at each range. (b) You need to resolve 5 mg and survive 12 g shocks without clipping. Which range do you configure, and what is the consequence for resolution?

Answers

2.1 Accuracy is closeness of a reading to the true value — a systematic property. **Precision** is the repeatability of readings about their own mean — a random-scatter property, which says nothing about truth. **Resolution** is the smallest change the output is able to represent, which is a property of the reporting scale and not of correctness at all. Example: a sensor with a large uncorrected offset but very low noise, such as the 1012.4 hPa device in §2.3.

2.2 Offset — yes, at the temperature at which you calibrated. **Random noise — no**; only averaging reduces it, and averaging costs bandwidth. **Temperature drift — no**; a single point captures a single temperature, so removing drift requires measuring temperature and applying a compensation model. **Quantisation — no**; it is fixed by the LSB, although dithering and averaging can recover some detail below one LSB.

2.3 (a) $150 \times \sqrt{20} = 150 \times 4.472 = 671$ µg ≈ **0.67 mg**. (b) $150 \times \sqrt{80} = 1342$ µg ≈ **1.34 mg** — exactly double, because the bandwidth quadrupled and noise follows its square root.

2.4 The worst-case excursion from the 25 °C calibration point is to 60 °C, so $\Delta T = 35$ °C. Drift = 0.3×35 = **10.5 mg**, which as a tilt is arcsin(0.0105) ≈ **0.60°**.

2.5 The resolution is roughly 140 times finer than the measurand while the drift alone exceeds the entire measurand, so the part resolves — very precisely — a number that is wrong; resolution is not accuracy.

2.6 (a) 0.05 % of 20 kPa = **10 Pa**. (b) Because the figure is guaranteed only at 25 °C, and no temperature coefficient is published, the error at any other temperature is not merely large — it is **unknown**. An unspecified coefficient is not a small error, it is an unbounded one, and you cannot put an unbounded term into a budget you have to sign.

2.7 Any board-level effect: cross-axis sensitivity after reflow soldering, package stress from PCB flex, thermal gradients across the assembly, mounting compliance, or noise picked up from your own power supply. The **integrator** must determine it, by measurement on the assembled unit.

2.8 Sensor P. Its ±0.25 % of 20 kPa = 50 Pa ≈ **5.1 mm**, comfortably inside the ±20 mm requirement and, critically, *guaranteed across the whole operating temperature range*. Sensor Q's ±0.05 % = 10 Pa ≈ 1.0 mm at 25 °C, but its behaviour at 5 °C is unknown because no temperature coefficient is specified. You cannot write a defensible error budget for Q at all. A total error band over the operating range beats a headline accuracy at a single temperature.

2.9 At 200 Hz: typ = $60 \times \sqrt{200} = 60 \times 14.14$ = **849 µg ≈ 0.85 mg**, which as a tilt is arcsin(0.00085) ≈ **0.049°**. Max = $100 \times \sqrt{200}$ = **1414 µg ≈ 1.41 mg**, or ≈ **0.081°**. Quadrupling the bandwidth from 50 Hz doubled the noise, as expected.

2.10 (a) ±2 % of 1000 mg is ±20 mg, so a reading of 1015 mg is **within specification** — it is a 1.5 % scale-factor error. (b) A **known-reference measurement**: with the axis aligned to gravity you know the true value is 1 g, so dividing by 1.015 removes the scale-factor error. What remains uncorrected is the *change* of sensitivity with temperature (±0.005 %/°C typ), the zero-g offset drift, and the noise — a single-point scale calibration fixes the gain at one temperature and nothing else.

2.11 (a) 14 bits gives 16 384 codes across the full span, so resolution = 2 × range / 16 384: **±2 g → 0.244 mg/LSB**; ±4 g → 0.488; ±8 g → 0.977; **±16 g → 1.953 mg/LSB**. (b) Surviving 12 g without clipping forces **±16 g**, giving 1.953 mg/LSB. Since quantisation contributes $1.953/\sqrt{12} = 0.56$ mg RMS, resolving 5 mg is still comfortable — so in this case the range requirement costs you nothing that matters. That is the point of doing the arithmetic rather than assuming the trade-off hurts.